AWS Lab: EC2 Status Checks Sweep & Hardware Degradation Triage

Duration : 30 Min.

Scenario

A routine monitoring sweep reports that EC2 instance i-0abcdef123456789 is failing status checks (StatusCheckFailed_System = 1, StatusCheckFailed_Instance = 0). The junior engineer attempts to reboot the instance from inside the OS, but the check continues to fail. The engineer must distinguish between a System Status check failure (underlying physical host issue) and an Instance Status check failure (guest OS crash), and apply the correct remediation: stopping and starting the instance to migrate it to a healthy physical host.

Input Details

The cloud environment has been pre-configured with the following parameters: - **Platform:** Amazon EC2 CloudWatch Status Checks - **Metrics:** `StatusCheckFailed_System` (Hardware/Host) vs `StatusCheckFailed_Instance` (Guest OS) - **Host Degradation Remediation:** Stop and Start instance to migrate hardware host

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (derived from your login email before the `@` symbol, e.g. `student-demo01`). All custom resources and cloud configurations should incorporate this convention: - **Resource Naming:** Prepend or append `<prefix>` to your created AWS resources (e.g. `<prefix>-resource`). - **Candidate Tag:** Add tag `CandidatePrefix` = `<prefix>` on all managed cloud resources.

Task Objectives

Perform the following technical objectives directly in the AWS Management Console or using the AWS CLI: 1. Query instance status check details using aws ec2 describe-instance-status. 2. Differentiate between StatusCheckFailed_System and StatusCheckFailed_Instance. 3. Explain why OS reboot fails to resolve hardware faults. 4. Execute stop-instances followed by start-instances to trigger hypervisor re-instantiation on new hardware. 5. Output findings to ~/status_check_triage.json.

Verification

Once completed, run the verification script to check your progress and receive your score. The verification system will inspect the live resources in the AWS cloud:

```
python run.py
```

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	Check parsing: correctly identify System status check failure vs Instance check	2 Marks
TC2	Root cause explanation: articulate that System check failures represent AWS underlying hardware faults	2 Marks
TC3	Remediation logic: explain why stop/start migrates the instance while in-guest reboot does not	2 Marks
TC4	CLI sequence: construct valid stop-instances and start-instances workflow	2 Marks
TC5	Post-check verification: verify both status checks return 'passed'	2 Marks

Total Score: 10 Marks

Important Notes

- Perform all actions directly in the AWS Management Console or using the AWS CLI.
- Ensure all resource configurations, tags, and naming conventions match the requirements.
- Do not terminate or alter pre-existing cloud infrastructure outside the scope of this assessment.